

Social Media Content Engine – Prototype Flow

The prototype can work as a simple **Source → Understand → Generate → Validate → Review → Export** pipeline.

1. User uploads the source

User does:

- Uploads a PDF/document or pastes text.
- Example: CERT-In advisory or CISA vulnerability report.

Technically:

PDF / DOCX / TXT / JSON



Document Parser



Text Extraction



Cleaned Source Text

For PDFs, we can use a PDF/text extraction library. If the document is scanned, OCR can be added later.

2. User selects what they want

User selects:

- LinkedIn Post
- X/Twitter Thread
- Or both

The user can also select:

- Target audience
- Tone
- Objective
- Language
- Detail level

Technically:

These become structured parameters sent to the backend:

Platform: LinkedIn + X

Audience: General Public

Tone: Professional

Objective: Inform

Language: English

Detail: Medium

3. Backend processes the source

Technically:

The backend sends the extracted text to the LLM.

For the prototype, we can use:

Frontend



Backend API



Ollama



Local LLM

Ollama allows us to run the selected open-source model locally.

4. AI first extracts the important information

Instead of directly asking the LLM:

"Write a LinkedIn post."

we first ask it to convert the source into a **structured representation of facts**.

For example:

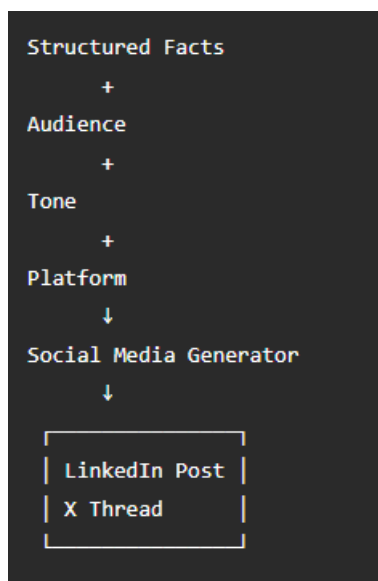
{

```
"title": "...",  
"date": "...",  
"vulnerability": "...",  
"affected_products": [...],  
"severity": "...",  
"impact": "...",  
"recommendations": [...],  
"source": "CERT-In Advisory"  
}
```

This becomes the **source-grounded knowledge layer / single source of truth** for the social-media outputs.

5. Generate the social-media content

The structured information + user's settings are passed to the generation module.



The prompt will explicitly tell the model:

- Use only information from the source.
- Do not invent facts.
- Adapt the content to the selected platform.
- Follow the selected audience and tone.

- Do not expose sensitive information.
-

6. Automatic validation

This is an important technical part of our prototype.

After generation, the output goes through multiple validators.

A. Fact / Claim Validation

Source Facts



Generated Claims



Compare



Supported / Unsupported

The system checks whether claims in the generated post are actually supported by the original source.

B. Sensitive Information Detection

The system checks for things such as:

- PII
- Internal IP addresses
- Credentials/secrets
- Internal hostnames
- Confidential incident information
- Other restricted information

Generated Content



Sensitive Information Detector



PASS / FLAG

C. Platform Validation

For X:

Generated Thread



Character Counter



Within Limit?



YES → Continue

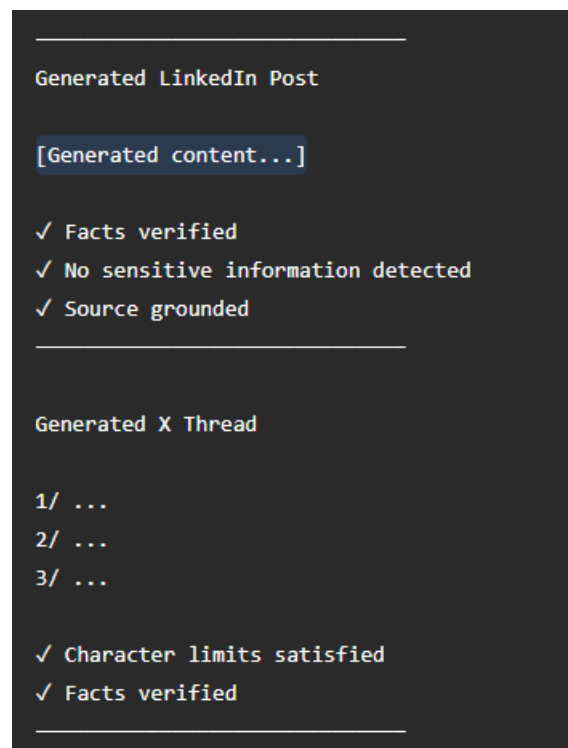
NO → Regenerate/Compress

D. Cross-Output Consistency

We can also compare the LinkedIn post and X thread against the same structured facts to make sure they don't contradict each other.

7. Human review

The user sees something like:



The user can:

Approve | Edit | Regenerate

This gives us a **human-in-the-loop** system rather than automatically publishing AI-generated government communication.

8. Export / final output

After approval:

Approved Content



Copy

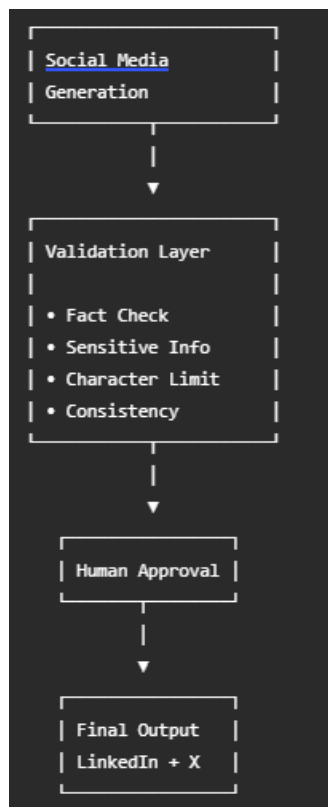
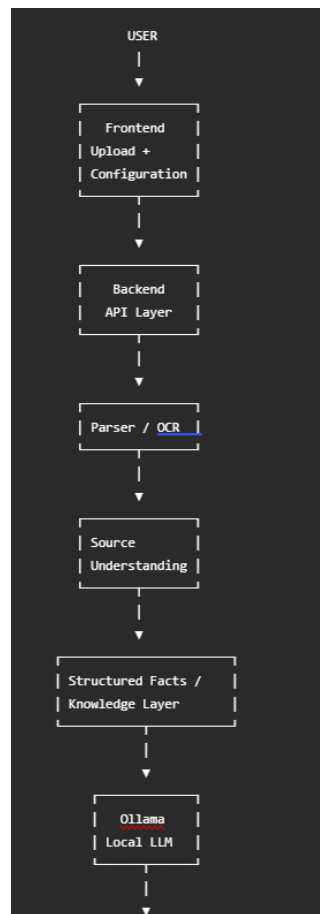
Download

Save

Publish*

For the first prototype, we can simply provide **Copy/Download**. Direct social-media publishing can be added later.

Overall Technical Architecture



Social Media Content Engine – UI Prototype

